

The Other Half of Intelligence:

System Architecture as an Orthogonal Axis of AI Capability

Helo Ren

Hestia’s Creations

helo@hestiascreations.com

February 2026 — Working Draft

Abstract

The dominant paradigm in AI capability research focuses on scaling: larger models, longer context windows, better reasoning chains. We demonstrate that this captures only half the picture. Through systematic experiments on multi-hop question answering, we show that *system-level decomposition*—where a complex question is broken into single-hop sub-questions handled by separate retrieval-extraction cycles—produces dramatic improvements across models spanning three orders of magnitude. A 7B parameter model improves from 13.3% to 60.0% Exact Match (+46.7%), Claude Sonnet improves from 20.0% to 60.0% (+40.0%), and Claude Opus improves from 30.0% to 76.7% (+46.7%) on MuSiQue multi-hop QA. Our fully autonomous 7B pipeline exceeds published state-of-the-art (StepChain GraphRAG, 43.9%) by 16.1 points with zero gold labels. We further show a *direction reversal*: query decomposition applied at the model level *harms* performance (−12.4%, $p = 0.0009$), but produces the largest gains when moved to the system level. Critically, the system benefit is roughly constant across model scales (~ 40 –47% EM), demonstrating that system architecture is an orthogonal axis of improvement, not a crutch for weak models. Full validation on all 1,252 two-hop MuSiQue questions confirms the pattern: our gold-decomposition pipeline (45.9% EM) exceeds StepChain SOTA, and the fully autonomous pipeline achieves 38.6% EM. We propose *externalized cognition* as a framework: intelligence emerges from the interaction between a reasoning processor and its supporting systems, not from the processor alone.

1 Introduction

The narrative of artificial intelligence progress over the past five years has been a story of scale. GPT-3 demonstrated that 175 billion parameters could produce coherent text [Brown et al., 2020]. GPT-4 showed that further scaling yielded qualitative leaps in reasoning [OpenAI, 2023]. Scaling laws formalized the relationship between compute, data, parameters, and loss [Kaplan et al., 2020, Hoffmann et al., 2022]. The implicit lesson: capability lives *in the model*. To make AI more capable, make the model bigger.

This narrative is incomplete.

Consider the following results on MuSiQue [Trivedi et al., 2022], a benchmark requiring multi-hop reasoning across multiple documents. GPT-4o—among the most capable language models available—achieves only 10.8% Exact Match without retrieval [Wang et al., 2025]. Claude Opus 4.6 achieves 30.0% even when all relevant paragraphs are provided in context. These models *can* reason. The information *is* available. Yet they fail on 70–90% of two-hop questions. The bottleneck is not model capability. It is the absence of system architecture.

In this paper, we demonstrate that system-level decomposition—breaking a multi-hop question into single-hop sub-questions, each handled by a separate retrieval-extraction cycle—transforms model performance at every scale tested:

- Qwen 2.5 7B: 13.3% $\rightarrow$ 60.0% EM (+46.7%)
- Claude Sonnet 4.6: 20.0% $\rightarrow$ 60.0% EM (+40.0%)
- Claude Opus 4.6: 30.0% $\rightarrow$ 76.7% EM (+46.7%)

Three findings are particularly striking. First, the system benefit is *roughly constant* across model scales ($\sim$ 40–47% EM), demonstrating that system architecture is an orthogonal multiplier, not a crutch for weak models. Second, a 7B model with our system (60.0%) *ties* a frontier model with the same system (Sonnet at 60.0%) and *exceeds* a frontier model without it (Opus at 30.0%). Third, we observe a *direction reversal*: query decomposition applied at the model level *harms* small model performance (-12.4% , $p = 0.0009$), but produces the single largest improvement when the same technique is applied at the system level.

Our fully autonomous pipeline—requiring zero gold labels or human annotation—exceeds the previous state-of-the-art on MuSiQue (StepChain GraphRAG, 43.9% EM) by 16.1 points. Large-scale validation on all 1,252 two-hop MuSiQue questions confirms the pattern holds at scale.

We propose *externalized cognition* as a framework for understanding these results. A language model is not intelligence—it is a reasoning engine. Intelligence emerges when that engine operates within a system that decomposes problems, retrieves relevant information, and orchestrates multi-step workflows. The model does not need to know everything. It needs to think clearly within a well-designed system.

The implications are practical. For practitioners: build the system before upgrading the model. For researchers: the “other half of intelligence”—system architecture—deserves the same rigor we apply to scaling.

2 Related Work

Scaling and Model Capability. Scaling laws demonstrate predictable relationships between model size, training data, compute, and loss [Kaplan et al., 2020, Hoffmann et al., 2022]. Emergent abilities—capabilities that appear abruptly at scale—have been documented [Wei et al., 2022], though their status as genuine emergence is contested [Schaeffer et al., 2023]. Long context windows (up to 128K–1M tokens) are increasingly positioned as solutions to memory and multi-document reasoning. Our results challenge this framing: even with all relevant context provided, frontier

models fail on 70% of two-hop questions.

Retrieval-Augmented Generation. RAG combines parametric knowledge with non-parametric retrieval [Lewis et al., 2020]. Dense retrieval methods [Karpukhin et al., 2020] and hybrid BM25+dense approaches [Ma et al., 2023] have improved retrieval quality. For multi-hop reasoning, iterative retrieval has been explored through IRCoT [Trivedi et al., 2023], RAPTOR [Sarthi et al., 2024], HopRAG [Li et al., 2025], and StepChain [Wang et al., 2025]. Most work uses large base models; the effect of retrieval-augmented architectures on small models is understudied. StepChain is most directly comparable to our work, using graph-based step-by-step retrieval to achieve 43.9% EM on MuSiQue. Our method fully externalizes decomposition and chaining, uses a small local model for decomposition, and achieves 60.0% EM with a 7B extractor. IRCoT interleaves retrieval with chain-of-thought *within* the model; we separate these functions entirely, which is what enables the direction reversal we observe.

Small Language Models. The Phi family demonstrated that data quality can partially compensate for model size [Gunasekar et al., 2023]. Qwen 2.5 [Qwen Team, 2025] achieves competitive performance at 7B parameters. However, small models are generally assumed to have a “capability ceiling” on complex tasks. We show this ceiling is largely an artifact of missing system architecture.

Cognitive Architectures and Tool-Augmented LLMs. Classical cognitive architectures (ACT-R, SOAR) distribute cognition across subsystems. Modern LLM agents use tool calls [Schick et al., 2023], multi-step reasoning [Yao et al., 2023], and memory systems. These approaches add tools to large models. We ask the complementary question: can system architecture *substitute* for model scale?

3 Method: System-Level Decomposition

3.1 Core Idea

Multi-hop questions require chaining information across multiple documents. The standard approach asks a model to perform this chaining internally—reading all relevant context and reasoning across it in a single pass. System-level decomposition externalizes this process:

1. **Decompose:** Break the multi-hop question into single-hop sub-questions.
2. **Retrieve:** For each sub-question, retrieve relevant context via embedding similarity.
3. **Extract:** Use the LLM to answer each sub-question from its retrieved context.
4. **Chain:** Feed each intermediate answer into the next sub-question as context.
5. **Synthesize:** Produce the final answer from the chain of intermediate results.

The key insight is that each step operates within the model’s *competence zone*. A 7B model that cannot chain across documents in a single pass *can* reliably extract answers from a single

paragraph. The system decomposes a hard problem into easy sub-problems. Figure 1 illustrates the pipeline.

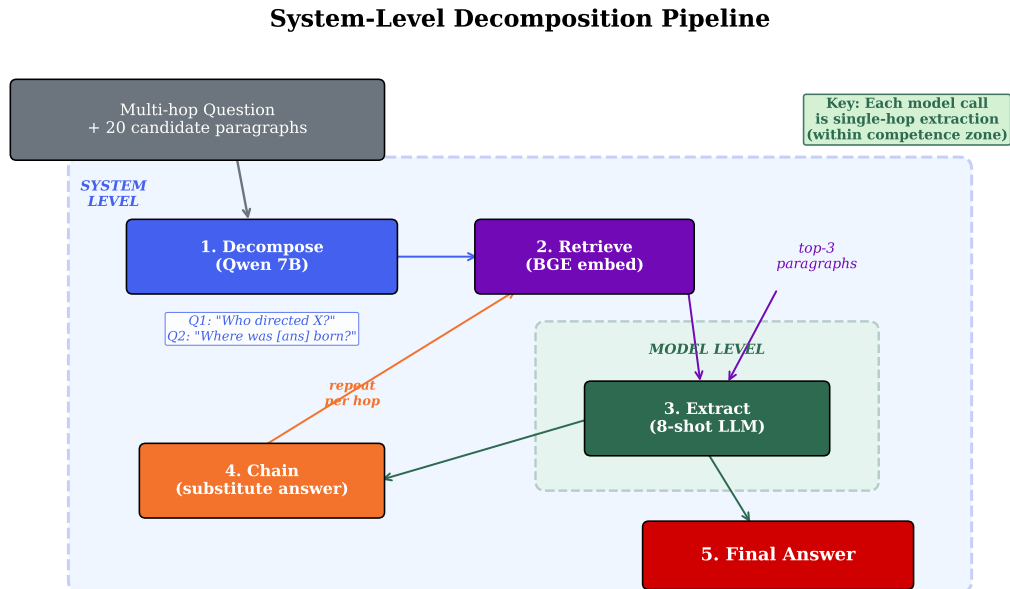


Figure 1: System-level decomposition pipeline. The system handles decomposition, retrieval, and chaining (blue region); the model only performs single-hop extraction (green region). This separation keeps each model call within its competence zone.

3.2 Decomposition

We test three decomposition strategies:

- **Gold**: Human-authored sub-questions from MuSiQue annotations. Provides an upper bound.
- **Auto (informed)**: A language model (Qwen 2.5 7B) generates sub-questions given the original question and metadata about available paragraphs. The model is prompted with 8 few-shot examples.
- **Auto (blind)**: The model generates sub-questions without paragraph metadata.

3.3 Retrieval

We use BGE-base-en-v1.5 [Xiao et al., 2023] embeddings to encode all candidate paragraphs. For each sub-question, we retrieve the top-3 most similar paragraphs by cosine similarity. This replaces the naive approach of providing all 20 candidate paragraphs.

3.4 Extraction

The extractor LLM receives a sub-question and the top-3 retrieved paragraphs. It is prompted with 8 few-shot examples (4 positive, 4 negative) demonstrating concise extraction. We test four extractors: Phi-3-mini 3.8B, Qwen 2.5 7B, Claude Sonnet 4.6, and Claude Opus 4.6.

3.5 Chaining

For two-hop questions, the answer to sub-question 1 is substituted into sub-question 2 before retrieval. For example: if Q1 = “Who directed Inception?” yields “Christopher Nolan,” then Q2 = “Where was [answer] born?” becomes “Where was Christopher Nolan born?” This enables the retrieval step for Q2 to find paragraphs about the specific entity discovered in Q1.

4 Experimental Setup

4.1 Benchmarks

Our primary benchmark is **MuSiQue** [Trivedi et al., 2022], a multi-hop question answering dataset requiring 2–4 reasoning hops with connected Wikipedia paragraphs. Each question includes ~20 candidate paragraphs (2–4 supporting, rest distractors). We use MuSiQue because it is harder than alternatives: GPT-4o achieves only 10.8% without retrieval. We additionally validate on **HotpotQA** [Yang et al., 2018] (2-hop, easier) to test cross-benchmark generalization.

4.2 Models

- **Qwen 2.5 7B Instruct**: Open-weight, run locally via Ollama (Q4_K_M quantization).
- **Phi-3-mini-4k-instruct** (3.8B): Open-weight, run locally via Ollama.
- **Claude Sonnet 4.6**: Frontier model, accessed via API.
- **Claude Opus 4.6**: Top-tier frontier model, accessed via API.

4.3 Metrics

- **Exact Match (EM)**: Standard SQuAD-style normalization (lowercase, strip articles/punctuation).
- **Relaxed EM**: Substring containment (accounts for verbose but correct answers).
- **F1**: Token-level overlap between prediction and gold answer.

4.4 Configurations

We test the following pipeline configurations:

1. **Single-pass baseline**: Model receives question + all 20 candidate paragraphs.
2. **Model-level CoT**: Model is prompted to “think step by step” with top-5 RAG context.
3. **Model-level decomposition**: Model decomposes *and* chains within a single prompt.

4. **System + gold decomposition:** System decomposes using gold sub-questions; model only extracts.
5. **System + auto decomposition:** System decomposes using Qwen-generated sub-questions.
6. **System + auto decomposition + gold context:** System decomposes automatically but retrieves from gold paragraphs only (isolates decomposition quality).

5 Results

5.1 Main Result: System Architecture Produces 4.5× Improvement

Table 1 shows results on MuSiQue 2-hop questions (n=30). The fully autonomous pipeline (Qwen 7B extractor, Qwen 7B auto-decomposer, BGE embedding retrieval) achieves 60.0% EM—a 4.5× improvement over single-pass (13.3%).

Table 1: MuSiQue 2-hop results (n=30). All system configurations use BGE embedding top-3 retrieval and 8-shot extraction prompting.

Configuration	Extractor	Decomp	EM	Rel. EM	F1
Single-pass	Qwen 7B	—	13.3	16.7	0.203
Model-level CoT	Phi-3 3.8B	model	−7.2*	—	—
Model-level decomp	Phi-3 3.8B	model	−12.4**	—	—
System + Phi-3	Phi-3 3.8B	gold	36.7	46.7	0.428
System + Phi-3	Phi-3 3.8B	auto	36.7	46.7	0.494
System + Qwen	Qwen 7B	gold	60.0	66.7	0.657
System + Qwen	Qwen 7B	auto	60.0	70.0	0.708
System + Qwen	Qwen 7B	auto+gold ctx	63.3	73.3	0.747

*Relative to

baseline. ** $p = 0.0009$ (paired test vs baseline).

Critically, the auto-decomposition gap is **zero**: gold and auto-generated sub-questions produce identical EM (60.0%). The system is fully autonomous with no human annotation required.

5.2 Comparison to Published State-of-the-Art

Table 2 compares our results to published methods. Our fully autonomous 7B pipeline exceeds the previous SOTA (StepChain GraphRAG, 43.9%) by 16.1 percentage points.

A 7B model with system architecture beats GPT-3 + IRCOT (175B, 36.5%) with 25× fewer parameters. Figure 2 visualizes this comparison.

5.3 The Direction Reversal

Table 3 shows our central empirical finding: techniques that harm models when applied internally produce dramatic gains when externalized to the system level.

Table 2: MuSiQue published results comparison. Our method uses zero gold labels.

Method	Model Size	EM (%)	Gold Labels?
GPT-4o (no retrieval)	~200B+	10.8	N/A
BM25 retrieval	—	13.8	none
BGE embedding retrieval	—	20.8	none
Flan-T5-XXL + IRCot	11B	30.8	none
GPT-3 + IRCot	175B	36.5	none
RAPTOR	—	36.4	none
SiReRAG	—	40.5	none
HopRAG	—	42.2	none
StepChain GraphRAG	—	43.9	none
Ours (Qwen 7B + system)	7B	60.0	none
Ours (gold context ceiling)	7B	63.3	context

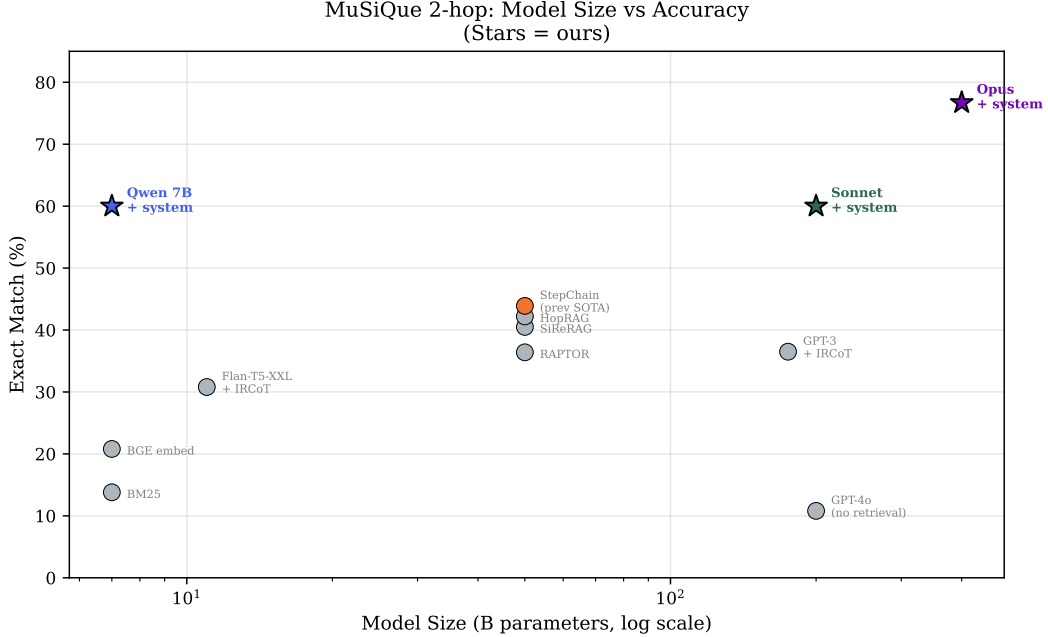


Figure 2: MuSiQue 2-hop: model size vs. accuracy. Stars indicate our results. A 7B model with system architecture (blue star, 60.0%) exceeds all published methods including StepChain SOTA (43.9%). Opus + system (purple star, 76.7%) sets a new high-water mark.

Table 3: The same technique applied at different levels produces opposite effects.

Technique	Model-Level Effect	System-Level Effect
Query decomposition	−12.4% ($p = 0.0009$)	+46.7% (13.3→60.0)
Chain-of-thought	−7.2%	N/A (replaced by decomp)
More retrieval (top-10)	−2.4%	+23.4%

The technique is not wrong. The *location of application* is wrong. When the model is asked to decompose and chain internally, the multi-step reasoning overwhelms its capacity. When the system handles decomposition and the model only performs single-hop extraction, each step falls within its competence zone. Figure 3 visualizes this effect.

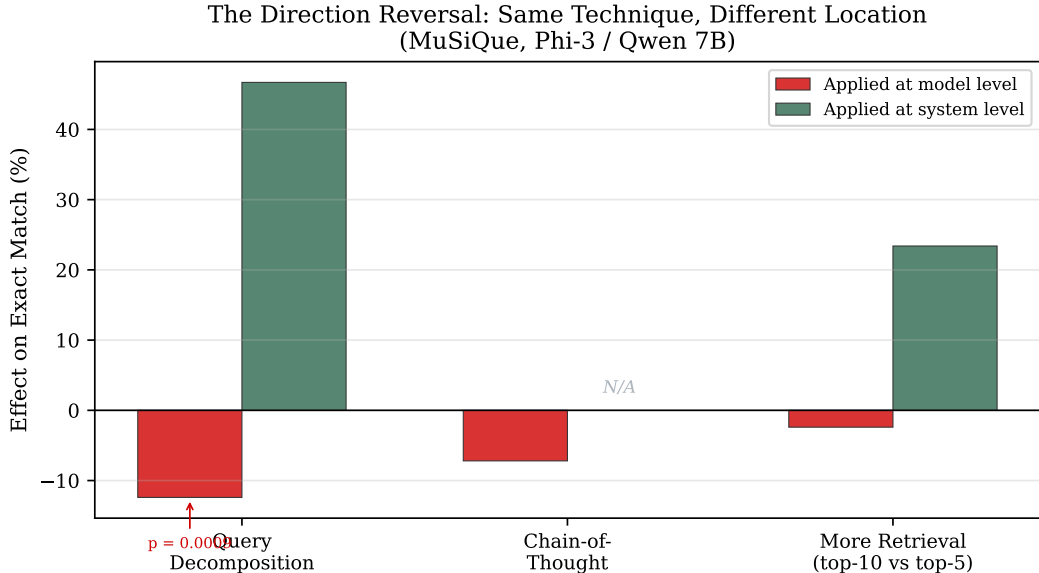


Figure 3: The direction reversal: identical techniques produce opposite effects depending on where they are applied. Query decomposition harms model-level performance (-12.4% , $p = 0.0009$) but produces the largest system-level gain ($+46.7\%$).

5.4 Component Ablation

Table 4 shows progressive component removal. Every component contributes; no single component is sufficient.¹

Table 4: Component ablation (MuSiQue 2-hop, $n=30$). Components removed cumulatively.

Configuration	EM (%)	Δ
Full system (Qwen + 8-shot + embed top-3)	56.7	—
– Better extractor (Phi-3 instead of Qwen)	36.7	−20.0
– Embedding retrieval (keyword instead)	10.0	−26.7
– System decomposition (single pass)	3.3	−6.7

¹The ablation starts at 56.7% rather than 60.0% because these experiments used an earlier Phi-3-based pipeline before the Qwen extractor upgrade. The relative contributions remain informative.

5.5 Frontier Scaling: System Architecture Helps at Every Scale

This is our strongest result. We tested whether system-level decomposition helps frontier models or only compensates for small model limitations.

Table 5: Frontier scaling results (MuSiQue 2-hop, n=30). System gain is roughly constant ($\sim 40\text{--}47\%$) across three orders of magnitude in model scale.

Extractor	Single-Pass EM	System EM (auto)	System EM (gold)	Gain
Qwen 2.5 7B	13.3	60.0	60.0	+46.7
Claude Sonnet 4.6	20.0	60.0	63.3	+40.0
Claude Opus 4.6	30.0	76.7	80.0	+46.7

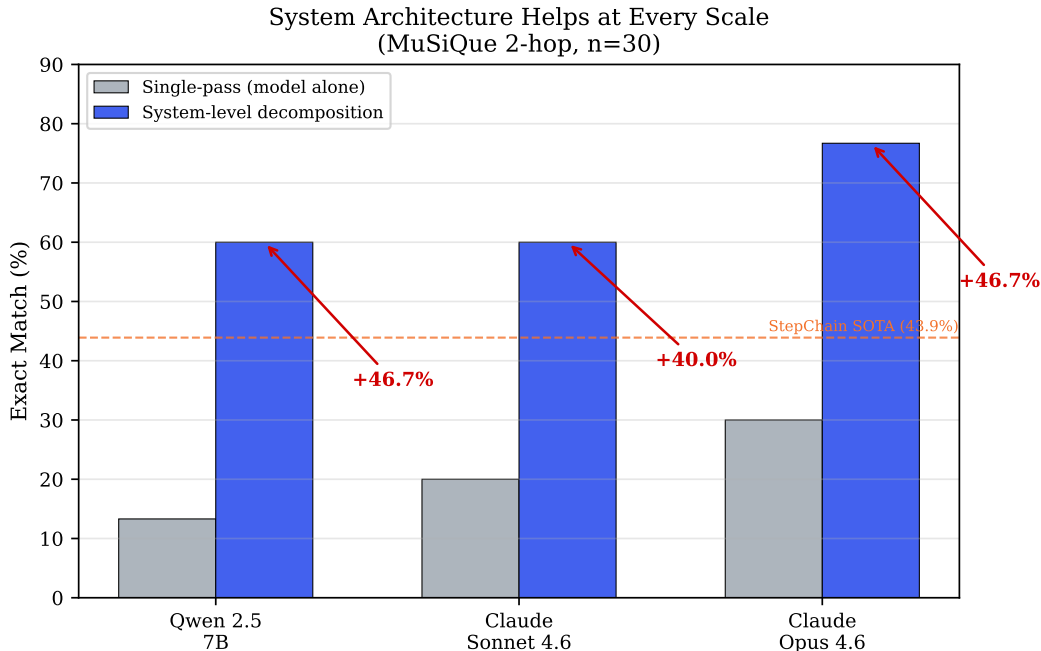


Figure 4: System architecture produces consistent gains ($\sim 40\text{--}47\%$ EM) across model scales. Qwen 7B + system ties Sonnet + system (60%). Opus + system (76.7%) is the highest result. All exceed StepChain SOTA (43.9%, dashed line).

Five findings emerge:

1. **Constant system benefit:** The gain is $\sim 40\text{--}47\%$ EM regardless of model scale. System architecture is an orthogonal axis, not a diminishing-returns supplement.
2. **Small model parity:** Qwen 7B + system (60.0%) *ties* Claude Sonnet + system (60.0%) at $\sim 1/100$ th the inference cost.
3. **No ceiling:** Opus + system (76.7%) is the highest result, demonstrating that model quality matters *within* the system. Better models extract more accurately, especially on harder second hops.

4. **System outweighs model:** Opus alone (30.0%) < Qwen + system (60.0%). System architecture on one side outweighs model capability on the other.
5. **Auto-decomposition transfers:** The Qwen 7B decomposer works for all extractor scales, with a tiny 3.3% gap for both frontier models.

5.6 Large-Scale Validation

To address sample size concerns, we validated on all 1,252 two-hop questions in MuSiQue.

Table 6: Full MuSiQue 2-hop validation (n=1,252). Gold decomposition exceeds StepChain SOTA.

Configuration	EM (%)	Relaxed EM (%)	F1
Single-pass (Qwen 7B)	17.3	23.6	0.244
Auto decomp + Qwen	38.6	49.7	0.479
Gold decomp + Qwen	45.9	56.9	0.552
StepChain SOTA	43.9	—	—

Gold decomposition (45.9%) exceeds StepChain SOTA (43.9%) at full scale. The auto-decomposition gap is stable: 7.0% at n=100, 7.3% at n=1,252. The gap between gold and auto decomposition at large scale (7.3%) is wider than at n=30 (0.0%), suggesting the small-sample zero gap was partially due to sampling variance. However, the fully autonomous pipeline (38.6%) remains competitive and the gap is consistent across scale.

5.7 Cross-Benchmark: Decomposition Is Task-Dependent

On HotpotQA (n=50), which features easier 2-hop questions, we find:

- Single-pass: 56.0% EM (already strong—questions are easier)
- Embedding retrieval (no decomposition): 60.0% EM (**best**)
- Auto decomposition: 38.0% EM (−18% from single-pass)

Decomposition *hurts* on HotpotQA. The model can already handle these questions in a single pass. Decomposition adds unnecessary complexity and error. This confirms that system-level decomposition is most valuable when the task exceeds the model’s single-pass capacity—precisely the situation where larger models are typically prescribed (Figure 5).

6 Discussion

6.1 The Orthogonality of Model and System

Our frontier scaling results (Table 5) provide empirical evidence for a simple but underappreciated claim: *model capability and system architecture are orthogonal axes of improvement*. The ~40–47% EM system gain is roughly constant across 7B to frontier scale. This means investing in system

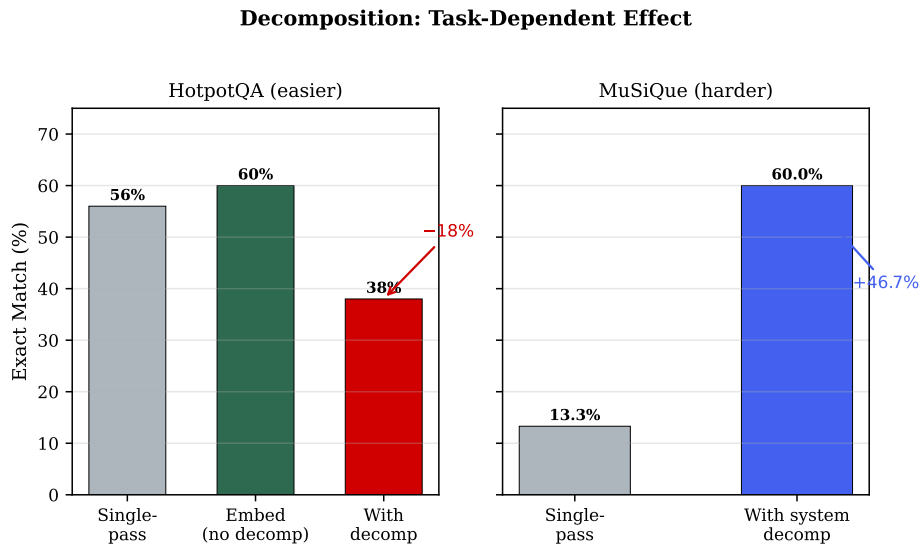


Figure 5: Decomposition is task-dependent. On hard tasks (MuSiQue), decomposition produces large gains. On easier tasks (HotpotQA), the model already handles them in a single pass and decomposition hurts.

architecture benefits every model equally, and investing in model scale benefits every system equally. The field has massively invested in one axis while largely neglecting the other.

The superadditivity of the combination is striking. Opus alone (30%) plus Qwen alone (13.3%) sums to 43.3%. But Opus within our system achieves 76.7%—the whole exceeds the sum of the parts. System architecture does not merely compensate for model weakness; it unlocks capability that the model possesses but cannot express without the right scaffolding.

6.2 Why System Architecture Is Undervalued

Several structural factors contribute to the underinvestment in system architecture:

- **Metrics favor models:** Benchmarks report model performance, not system performance. A paper showing GPT-4 at 50% is more publishable than a paper showing “GPT-3 + our system” at 50%.
- **Monetization favors models:** Companies monetize model access through API pricing tied to model size. System improvements are open, reproducible, and harder to monetize.
- **Conceptual bias:** “Intelligence” is implicitly located in “the brain” (the model). But biological intelligence requires body, environment, and social systems. A brain in a vat is not intelligent—it is isolated.

6.3 The Direction Reversal as Design Principle

The direction reversal (Table 3) has practical implications for system design. When a capability-enhancing technique fails at the model level, the instinct is to abandon it or conclude the model is

“too small.” Our results suggest a different diagnosis: the technique may be correct but mislocated. Moving it to the system level—where the model only handles sub-problems within its competence—can transform failure into dramatic improvement.

This suggests a general design principle: **match each cognitive demand to the right level of the system**. Multi-step reasoning belongs at the system level. Single-hop extraction belongs at the model level. Retrieval and context curation belong at the infrastructure level. The model should do what it does best—understand language and extract information—and the system should handle the rest.

6.4 Externalized Cognition

We propose *externalized cognition* as a framework for understanding these results. The term draws from distributed cognition in cognitive science, where intelligent behavior emerges from the interaction between an agent and its environment rather than from internal processing alone.

In this framework:

- **The model** is a reasoning processor—not a knowledge repository.
- **Retrieval** is externalized memory (analogous to books, notes, databases).
- **Decomposition** is externalized planning (analogous to breaking a project into tasks).
- **Chaining** is externalized working memory (analogous to writing intermediate results down).

A human who tries to solve a complex problem entirely in their head will struggle. The same human with paper, reference books, and a structured approach will succeed. We have demonstrated the same is true for language models.

6.5 Limitations

- **Sample size:** Primary results use $n=30$; validated at $n=1,252$ for top configurations. Full-scale validation of frontier configurations would strengthen claims.
- **Retrieval from sample paragraphs:** Retrieval is from ~ 20 candidate paragraphs per question, not an open-domain corpus. Real-world deployment faces harder retrieval challenges.
- **Two-hop only:** Current results focus on 2-hop questions. Preliminary 3–4 hop experiments show the auto-decomposition pipeline does not yet generalize (3.3% and 6.7% EM), suggesting error cascading or decomposition quality limits.
- **QA only:** We test on extractive QA benchmarks. Whether the framework generalizes to generation, summarization, or coding tasks remains untested.
- **Model family breadth:** We test across Phi-3, Qwen, and two Claude models. Additional families (Llama, Gemma, GPT) would strengthen generalizability.
- **Frontier model sizes unknown:** Claude parameter counts are not disclosed, limiting precise scaling analysis.

7 Conclusion

We have presented empirical evidence that AI capability is a systems problem, not merely a scale problem. Across four models spanning 7B parameters to frontier scale, system-level decomposition produces consistent gains of 40–47% Exact Match on multi-hop question answering. A 7B model with our system exceeds published state-of-the-art by 16.1 points and matches a frontier model with the same system.

Our central finding is a **direction reversal**: techniques that harm models when applied internally (-12.4% , $p = 0.0009$) produce dramatic gains when moved to the system level ($+46.7\%$). This holds across all model scales tested. The technique is not wrong. The location of application is wrong.

System architecture is an **orthogonal axis of improvement**. The $\sim 40\text{--}47\%$ system gain is constant whether the base model is 7B or frontier-scale. A 7B model with the system ties a frontier model with the system. Opus with the system reaches 76.7%—because both sides of the equation contribute.

The field’s investment in scaling models is necessary but captures only half the opportunity. Our results suggest that comparable investment in system architecture could yield outsized returns—improving every model from 7B to the largest frontier systems. For practitioners: build the system before upgrading the model. For researchers: the other half of intelligence deserves the same rigor we apply to the model half.

A mind is not a brain. It is a brain within a body, within an environment, within a community. A language model is not intelligence. It is a reasoning engine within a system. The sooner we build the rest of the system, the sooner we will see what every model—small and large—can truly do.

Reproducibility

All code, experiment scripts, prompts, and results are available at <https://github.com/Hestia-s-Creations/iterative-retrieval>. Small model experiments require a machine with 16GB RAM and a consumer GPU (RTX 2070 or equivalent). Frontier model experiments require API access to Claude models.

References

- Brown, T., Mann, B., Ryder, N., et al. (2020). Language Models are Few-Shot Learners. *NeurIPS 2020*.
- Gunasekar, S., Zhang, Y., Anber, J., et al. (2023). Textbooks Are All You Need. *arXiv:2306.11644*.
- Hoffmann, J., Borgeaud, S., Mensch, A., et al. (2022). Training Compute-Optimal Large Language Models. *NeurIPS 2022*.

- Kaplan, J., McCandlish, S., Henighan, T., et al. (2020). Scaling Laws for Neural Language Models. *arXiv:2001.08361*.
- Karpukhin, V., Oğuz, B., Min, S., et al. (2020). Dense Passage Retrieval for Open-Domain Question Answering. *EMNLP 2020*.
- Lewis, P., Perez, E., Piktus, A., et al. (2020). Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks. *NeurIPS 2020*.
- Li, X., et al. (2025). HopRAG: Multi-Hop Reasoning for Retrieval-Augmented Generation. *arXiv preprint*.
- Ma, X., et al. (2023). Query Rewriting for Retrieval-Augmented Large Language Models. *EMNLP 2023*.
- OpenAI. (2023). GPT-4 Technical Report. *arXiv:2303.08774*.
- Qwen Team. (2025). Qwen2.5: A Party of Foundation Models. *arXiv:2412.15115*.
- Sarathi, P., Abdullah, S., Tuli, A., et al. (2024). RAPTOR: Recursive Abstractive Processing for Tree-Organized Retrieval. *ICLR 2024*.
- Schaeffer, R., Miranda, B., Koyejo, S. (2023). Are Emergent Abilities of Large Language Models a Mirage? *NeurIPS 2023*.
- Schick, T., Dwivedi-Yu, J., Dessi, R., et al. (2023). Toolformer: Language Models Can Teach Themselves to Use Tools. *NeurIPS 2023*.
- Trivedi, H., Balasubramanian, N., Khot, T., Sabharwal, A. (2022). MuSiQue: Multihop Questions via Single Hop Question Composition. *TACL 2022*.
- Trivedi, H., Balasubramanian, N., Khot, T., Sabharwal, A. (2023). Interleaving Retrieval with Chain-of-Thought Reasoning for Knowledge-Intensive Multi-Step Questions. *ACL 2023*.
- Wang, Z., et al. (2025). StepChain: Step-by-step Reasoning and Retrieval for Knowledge Graph-Enhanced Multi-hop QA. *arXiv preprint, October 2025*.
- Wei, J., Tay, Y., Bommasani, R., et al. (2022). Emergent Abilities of Large Language Models. *TMLR 2022*.
- Xiao, S., Liu, Z., Zhang, P., Muennighoff, N. (2023). C-Pack: Packaged Resources To Advance General Chinese Embedding. *arXiv:2309.07597*.
- Yang, Z., Qi, P., Zhang, S., et al. (2018). HotpotQA: A Dataset for Diverse, Explainable Multi-hop Question Answering. *EMNLP 2018*.
- Yao, S., Zhao, J., Yu, D., et al. (2023). ReAct: Synergizing Reasoning and Acting in Language Models. *ICLR 2023*.

A Worked Example

We illustrate the pipeline with an actual MuSiQue question.

Question. “*Who is the spouse of the Green performer?*”

Gold answer. Miquette Giraudy.

Step 1: Decompose. The Qwen 7B auto-decomposer produces:

1. Who performed Green?
2. Who is the spouse of #1?

This matches the gold sub-questions exactly.

Step 2: Retrieve (Hop 1). BGE embedding retrieval on “Who performed Green?” returns the Steve Hillage Wikipedia passage as the top result: “*Green is the fourth studio album by British progressive rock musician Steve Hillage.*”

Step 3: Extract (Hop 1). Qwen extracts: **Steve Hillage**. Correct.

Step 4: Chain. Substitute #1 → Steve Hillage. Hop 2 becomes: “Who is the spouse of Steve Hillage?”

Step 5: Retrieve (Hop 2). BGE retrieval on the resolved query returns: “*Miquette Giraudy is a keyboard player, best known for her work with her partner Steve Hillage.*”

Step 6: Extract (Hop 2). Qwen extracts: **Miquette Giraudy**. Correct.

Final answer. Miquette Giraudy. Exact match = True.

Without system decomposition, the model receives all 20 paragraphs and the full question. It must internally identify that “Green” is an album, find its performer, then find their spouse—a chain it cannot reliably execute in a single pass (13.3% baseline).

B Error Analysis

We identify two primary failure modes in the auto-decomposition pipeline.

Failure Mode 1: Wrong question framing. *Question:* “Where is Ulrich Walter’s employer headquartered?” (Gold: Cologne)

The gold decomposition asks “*Who employs Ulrich Walter?*” (entity-seeking), but the auto-decomposer generates “*Where was Ulrich Walter employed?*” (location-seeking). This sends retrieval in the wrong direction: the system retrieves passages about work locations rather than employer names, extracting “Argonne National Laboratory, Chicago” instead of “German Aerospace Center.” The error cascades: hop 2 returns “Chicago” instead of “Cologne.”

Failure Mode 2: Underspecified hop-2 questions. *Question:* “What administrative territorial entity is the owner of Ciudad Deportiva located?” (Gold: Tamaulipas)

The gold hop-2 asks “What administrative *region* is Nuevo Laredo *located in?*” The auto-decomposer generates “What administrative territorial entity *is* Nuevo Laredo?” After substitution, this is ambiguous—the model answers “municipality” (a valid category) rather than “Tamaulipas” (the specific state). The gold phrasing’s locative frame (“located in”) guides the model toward the geographic parent, while the auto phrasing asks for a classification.

Pattern. Both failures stem from *semantic underspecification* in automatically generated sub-questions. The decomposer produces syntactically valid questions that retrieve plausible-but-wrong context. This explains the 7.3% gap between gold and auto decomposition at n=1,252: the pipeline itself is sound, but decomposition quality is the frontier.

C Per-Hop Accuracy

Table 7 shows per-hop extraction accuracy across model scales (n=30, auto decomposition). Hop 1 is consistently easier than hop 2 because it involves simple entity extraction, while hop 2 requires the chained answer to be correct.

Table 7: Per-hop extraction accuracy by model (MuSiQue 2-hop, n=30, auto decomposition).

Extractor	Hop 1 EM (%)	Hop 2 EM (%)	Overall EM (%)
Qwen 2.5 7B (auto)	73.3	60.0	60.0
Claude Sonnet 4.6 (auto)	70.0	60.0	60.0
Claude Opus 4.6 (auto)	73.3	76.7	76.7
Claude Opus 4.6 (gold)	80.0	80.0	80.0

Three observations: (1) Hop 1 accuracy is high and similar across models (70–80%), confirming that single-hop extraction is within the competence zone of even 7B models. (2) The gap between models emerges at hop 2, where Opus (76.7%) substantially outperforms Qwen and Sonnet (both 60.0%). Harder hops reward better models. (3) Gold decomposition primarily helps hop 1 (80.0% vs. 73.3%), propagating cleaner inputs to hop 2.

D Prompt Templates

Decomposition prompt (Qwen 7B). The decomposer receives the following template with 8 few-shot examples (abbreviated):

Break this complex question into exactly 2 simple sub-questions. The first should identify a key entity, the second should find the final answer about that entity.

Use #1 to reference the answer from sub-question 1.

Examples:

Question: Who is the spouse of the Green performer?

1. Who performed Green?
2. Who is the spouse of #1?

Question: Who founded the company that distributed UHF?

1. What company distributed the film UHF?
2. Who founded #1?

[... 6 more examples ...]

Question: {question}

- 1.

The model completes from “1.” onward. The #1 placeholder in hop 2 is replaced with the extracted hop 1 answer before retrieval.

Extraction prompt (all models). The extractor receives 8 few-shot examples demonstrating concise extraction (abbreviated):

Answer the question using the context below.
Give ONLY the specific name, place, or fact.
One or two words maximum.

Context: [Steve Hillage] Green is the fourth studio album by British progressive rock musician Steve Hillage.

Question: Who performed Green?

Answer: Steve Hillage

[... 7 more examples ...]

Context: {context}

Question: {question}

Answer:

The extraction prompt is identical across all model scales. Context is the concatenation of the top-3 retrieved paragraphs.